

Jack Vacek

(832) 593-2451 | vacekjack@gmail.com | linkedin.com/in/jackvacek/ | github.com/JackVacek

EDUCATION

Texas A&M University | Craig and Galen Brown Engineering Honors **College Station, TX**
Bachelor of Science in Computer Science | Emphasis: Japanese Language & Culture **September 2021 - May 2025**
Cumulative GPA: 3.97 | Honors: National Hispanic Scholar, Fall 2022 Dean's Honor Roll, and IAP Visa Scholarship

EXPERIENCE

MRC Global | Cybersecurity Intern, Houston, Texas **May 2024 – August 2024**

- Engineered automation tools for regulatory compliance reporting, resulting in **\$70,000+** cost savings and **14-day** reduction in quarterly workflow processes.
- Led implementation of enterprise-wide phishing awareness campaigns targeting **3,000+** employees using **20+** customized security templates.
- Developed department-based employee segmentation algorithm that increased phishing detection metrics by **1.5%**.
- Implemented Insider Risk Management protocols that successfully identified and mitigated **2+** cases of data theft attempts.

AI4ALL Ignite | Student, Stanford, California **January 2024 – August 2024**

- Completed comprehensive **13-week** artificial intelligence curriculum culminating in development and presentation of AI-driven solutions.
- Participated in **7-week** professional development program with industry mentors focusing on career preparation and network building in the AI sector.

PROJECTS

American Go: 3D Passenger Flight Visualization Platform | TypeScript, React, Next.js, Tailwind CSS, Google Maps API

- Engineered an interactive travel platform leveraging Google's 3D Maps API that transformed flight data from American Airlines' API into immersive visualizations with real-time rendering capabilities.
- Optimized performance by developing a selective flight rendering system that reduced memory consumption by **40%** while maintaining **sub-100ms** latency for **200+** simultaneous flights.
- Created an AI-powered itinerary generator using OpenAI that produced personalized travel recommendations with **95%** relevance based on destination and travel dates.

Kung Fu Tea Point-Of-Sale System | PostgreSQL, Java, JavaFX, Python, AWS, Express.js, React.js, ChakraUI

- Utilized PostgreSQL and Express.js to integrate AWS-hosted databases with a React.js frontend, enabling a seamless, responsive backend that processed **500+** daily transactions with **99.9%** reliability.
- Developed a desktop Point-of-Sale system using Java and JavaFX that increased order processing speed by **40%**.
- Implemented secure AWS cloud infrastructure supporting real-time inventory tracking and reporting capabilities for management decision-making.

RELEVANT COURSES

Advanced Technical: Competitive Programming, Parallel Computing, Cloud Computing, Machine Learning, Software Reverse Engineering, Human Computer Interaction, and Senior Capstone

Computer Science Core: Programming Languages, Design and Analysis of Algorithms, Discrete Structures for Computing, Computer Systems, and Program Design & Concepts

Honors Program: Data Structures & Algorithms, Computer Organization, Programming Studio, Linear Algebra

SKILLS

Programming Languages: Python, Java, JavaScript, TypeScript, C, C++, HTML, CSS

Web Development: React.js, Express.js, JavaFX, ChakraUI, RESTful APIs

Data Science: PyTorch, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn

Cloud & Infrastructure Services: AWS EC2, AWS Lightsail, Microsoft Azure, Docker, PostgreSQL

Development Tools: Git, GitHub, Jira, Jupyter Notebooks, Google Colab, UML

AFFILIATIONS

Aggie Competitive Programming, Texas A&M Cybersecurity Club, Japanese Culture Association